

Network Security 1 Project Rubric 6

Utilize Network Security Tools

Provide documentation of network security tools usage, including at least 1 Wireshark capture with analysis, 1 network vulnerability scanner report, and 1 network penetration testing tool output.

Utilize Network Security Tools — Technical Architecture

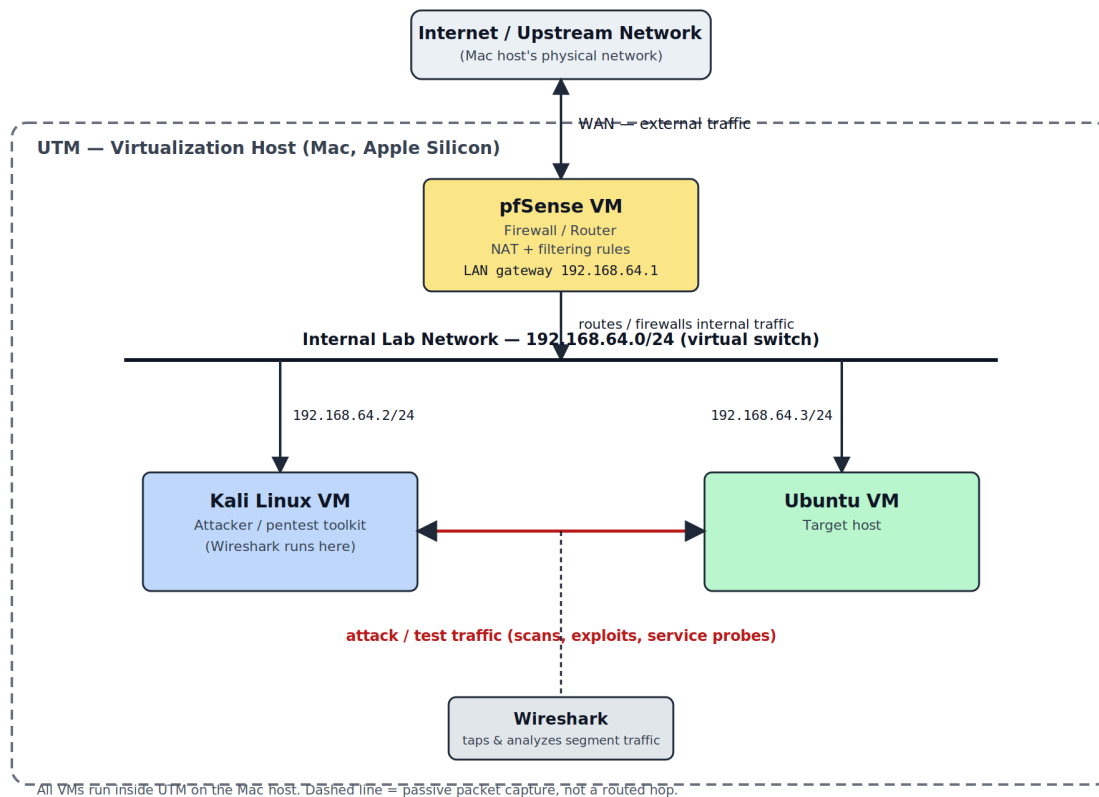
Assignment Requirement

Provide documentation of network security tools usage, including:

1. **One Wireshark capture with analysis**
2. **One network vulnerability scanner report**
3. **One network penetration testing tool output**

1. Technical Network Security Architecture

Network Security 1 — Rubric 6 Lab Topology



2. Tool #1 — Wireshark Packet Capture & Analysis

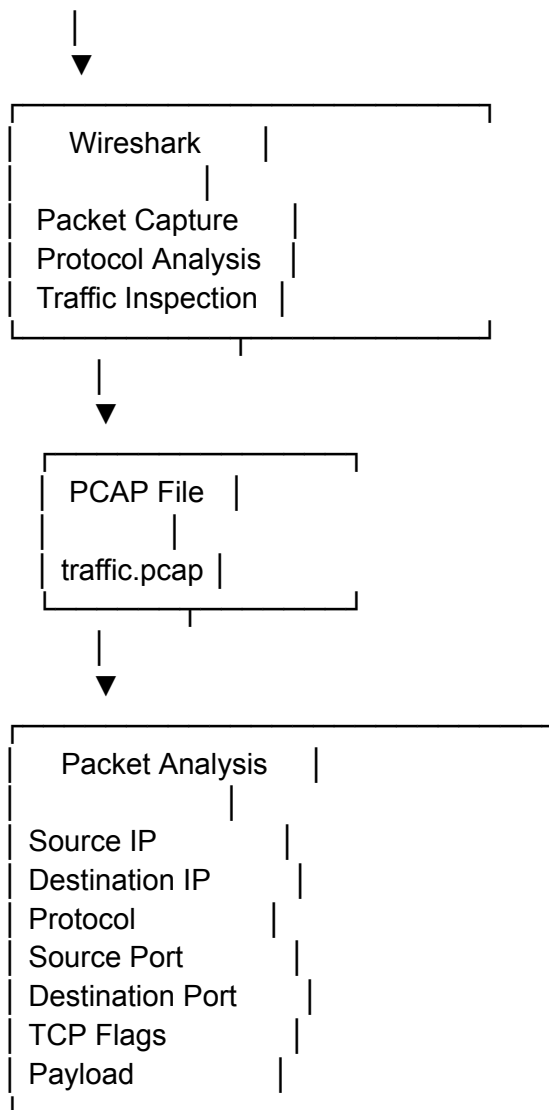
Purpose

Wireshark is used to capture and analyze network traffic traveling between systems.

Architecture

Kali Linux Analyst VM

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Documentation

Record:

Field	Example
Tool	Wireshark
Capture Interface	Lab Network Interface
Source	Kali Linux
Destination	Metasploitable

Capture Format	PCAP
Protocol	TCP
Source IP	192.168.10.20
Destination IP	192.168.20.30
Destination Port	80 / 21 / 22
Analysis	Identify normal or suspicious traffic

Analysis

The Wireshark analysis should identify:

- Source IP address
- Destination IP address
- Protocol
- Port number
- TCP flags
- Packet sequence
- Connection attempts
- Suspicious or unusual traffic
- Evidence of scanning or exploitation

Example Finding

Kali Analyst
192.168.10.20
|
| TCP SYN
| Port 80
▼

Metasploitable
192.168.20.30

Wireshark Analysis:

Source IP: 192.168.10.20
Destination IP: 192.168.20.30
Protocol: TCP
Destination: Port 80
TCP Flag: SYN

Finding:

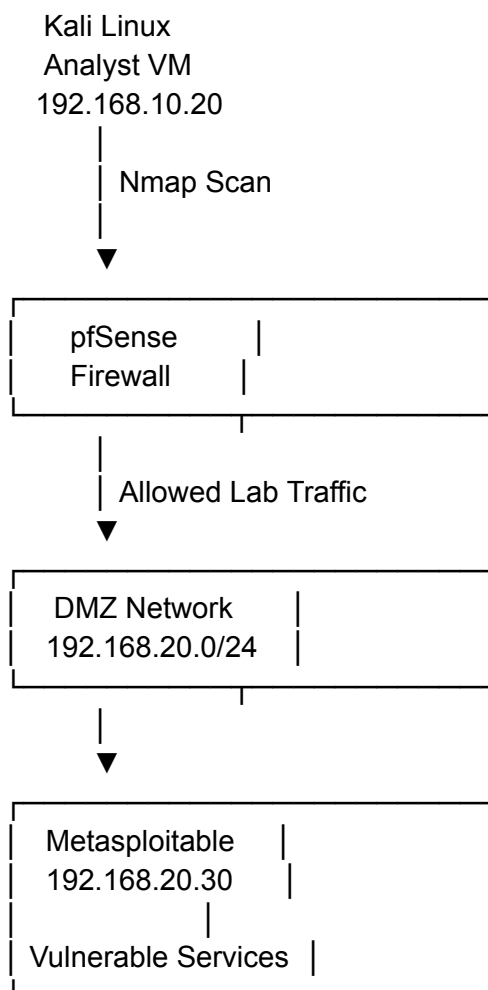
TCP connection attempt identified between the analyst system and the vulnerable system.

3. Tool #2 — Network Vulnerability Scanner

Tool: Nmap

Nmap is used to discover hosts, identify open ports and services, and perform network reconnaissance.

Architecture



Nmap Documentation

Record:

Field	Example
Tool	Nmap
Scanner	Kali Linux
Source IP	192.168.10.20
Target	192.168.20.30
Scan Type	TCP SYN Scan
Purpose	Identify open ports/services
Output	Nmap scan report

Example Output

Nmap scan report for 192.168.20.30

PORT	STATE	SERVICE
21/tcp	open	ftp
22/tcp	open	ssh
23/tcp	open	telnet
80/tcp	open	http
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds

Vulnerability Analysis

The scan results should be reviewed to determine:

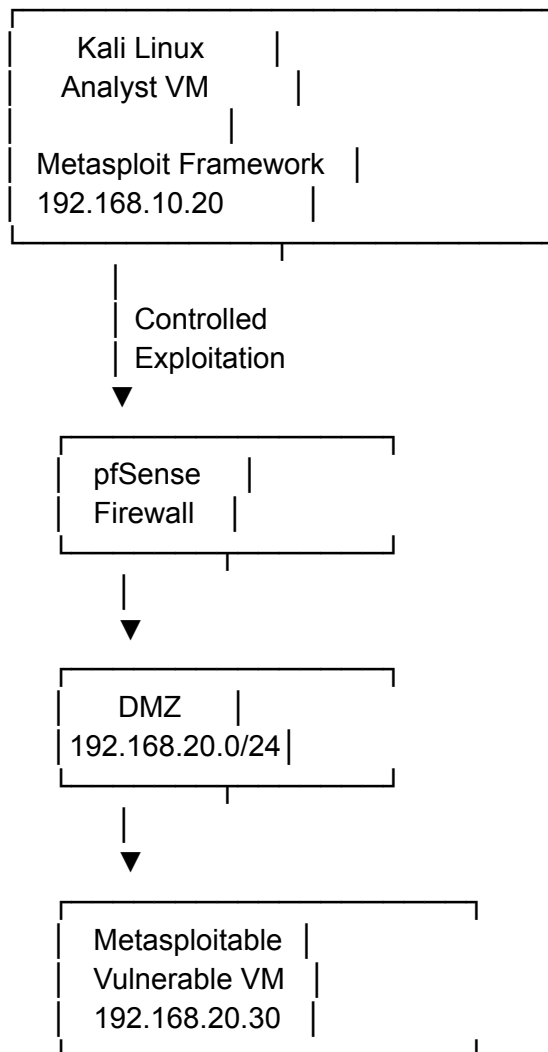
- Which ports are open?
 - Which services are running?
 - Which services are unnecessary?
 - Which services may be vulnerable?
 - Which vulnerabilities should be prioritized?
-

4. Tool #3 — Penetration Testing Tool

Tool: Metasploit Framework

Metasploit can be used in the controlled lab to validate whether a known vulnerability can actually be exploited.

Architecture



Documentation

Record:

Field	Example
Tool	Metasploit Framework
Attacker/Analyst	Kali Linux
Source IP	192.168.10.20
Target IP	192.168.20.30
Target	Metasploitable
Vulnerability	Documented vulnerable service
Exploit Module	Selected Metasploit module
Result	Successful / unsuccessful
Evidence	Console output

Example Output Structure

```
msf6 > use [authorized lab exploit module]
```

```
msf6 > set RHOSTS 192.168.20.30
```

```
msf6 > set RPORT [target port]
```

```
msf6 > check
```

```
[*] Target appears vulnerable
```

```
msf6 > run
```

```
[*] Exploit attempt initiated
```

```
[*] Target response received
```

Result:

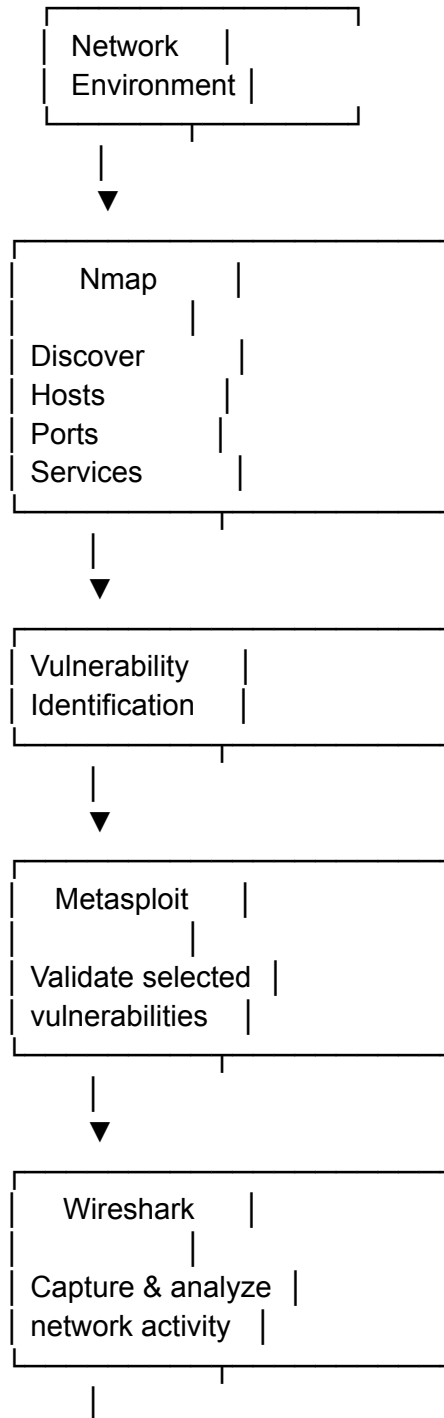
Controlled penetration test completed.

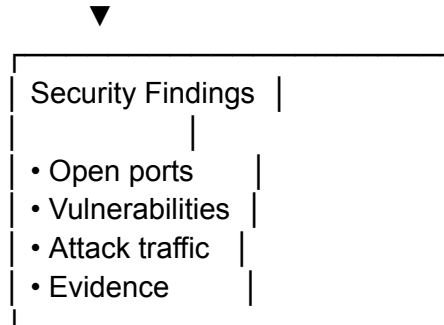
Important: Use this only against your intentionally vulnerable lab machine, such as Metasploitable.

5. How the Three Tools Work Together

This is the most important part of the architecture.

SECURITY ASSESSMENT WORKFLOW





6. Final Report Structure

For your assignment, I recommend organizing the report as:

1. Objective

Explain the purpose of using network security tools to identify, analyze, and validate security weaknesses.

2. Network Architecture

Show the lab network:

Kali → pfSense → DMZ → Metasploitable

3. Wireshark Capture

- Capture configuration
- Capture information
- Protocol analysis
- Important packets
- Findings
- Screenshot of capture
- PCAP evidence

4. Vulnerability Scanner — Nmap

- Target
- Scan type
- Open ports
- Services
- Findings
- Nmap output

- Vulnerability assessment

5. Penetration Testing — Metasploit

- Target
- Vulnerability tested
- Module/tool used
- Test procedure
- Output
- Result
- Evidence

6. Correlation of Findings

Explain how:

Nmap identifies → Metasploit validates → Wireshark observes

7. Security Recommendations

Based on the findings:

- Disable unnecessary services.
- Restrict unnecessary ports.
- Patch vulnerable services.
- Apply firewall restrictions.
- Segment vulnerable systems.
- Continue monitoring network traffic.

8. Conclusion

The three tools provide complementary security capabilities: **Nmap identifies the attack surface, Metasploit validates vulnerabilities in the authorized lab, and Wireshark provides packet-level evidence of the resulting network activity.**